Outline

- Classes of materials
- Chemistry review
- Review of atomic structure
- Review of the periodic table
- Density, atomic # & wt, mole, Avogadros #
- Bonding forces and energy

Classes of materials

	POLYMERS	CERAMICS	METALS
DUCTILITY	Varies	Poor	Good
CONDUCTIVITY (ELECTRICAL & THERMAL)	Low	Low	……..
HARDNESS/STRENGTH	Low – medium	Very high	Medium – high
CORROSION RESISTANCE	Fair – good	……	Fair – poor
STIFFNESS	Low	High	Fair
FRACTURE TOUGHNESS	Low – medium	Low	……
MACHINABILITY	Good	Poor	Good

Why study bonding?

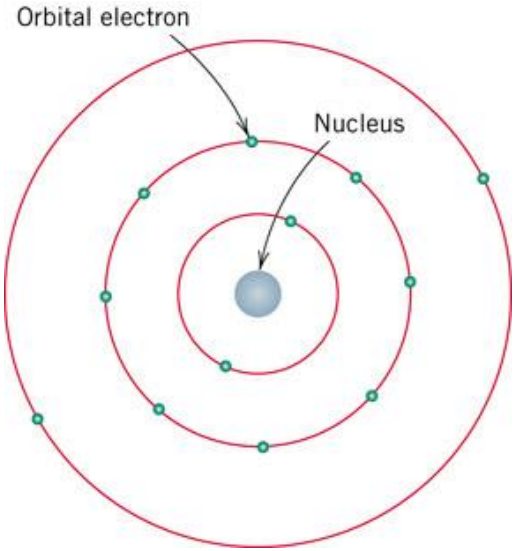
- Because the properties of materials (*strength, hardness, conductivity, etc..*) are determined by the manner in which atoms are connected.
- Also by how the atoms are arranged in space → …….

What determines the nature of the chemical bond between atoms?

- Electronic structure (distribution of electrons in atomic orbitals)
- Number of electrons and …… (tendency for an atom to attract an electron)

Atomic Structure

Bohr model of the atom: (1913)

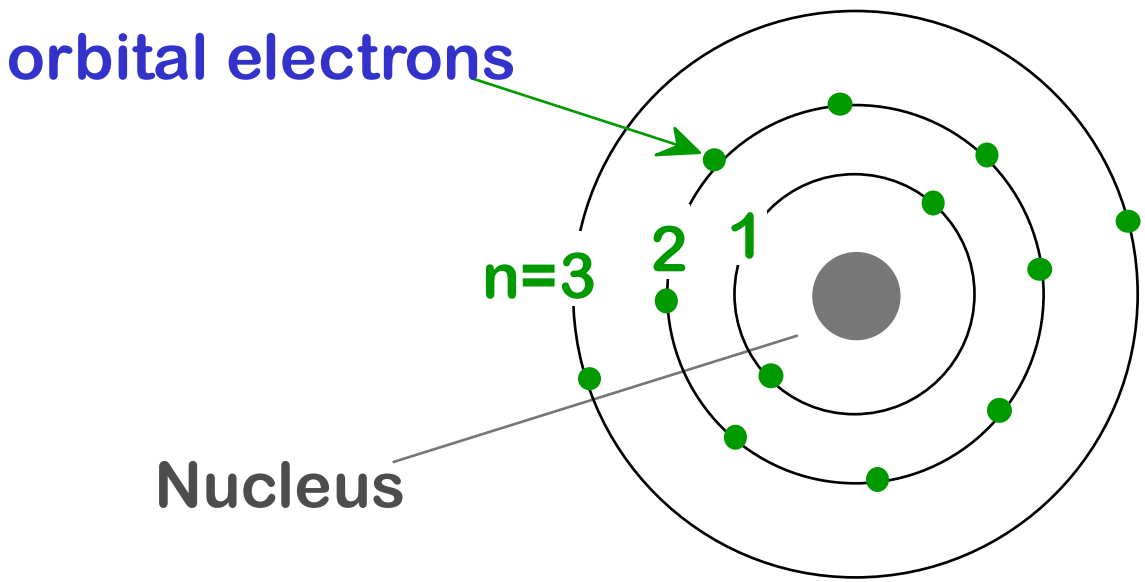


– Bohr atomic model

- electrons revolve around nucleus in discrete orbitals

Based on earlier work of Rutherford and his own from spectral emission studies

Bohr Atom



Z = # protons

= *1 for hydrogen to 94 for plutonium*

N = # neutrons

Atomic mass A ≈ Z + N

Isotopes: same Z, different N → same element, different atomic mass (e.g. ^{12}C and ^{13}C).

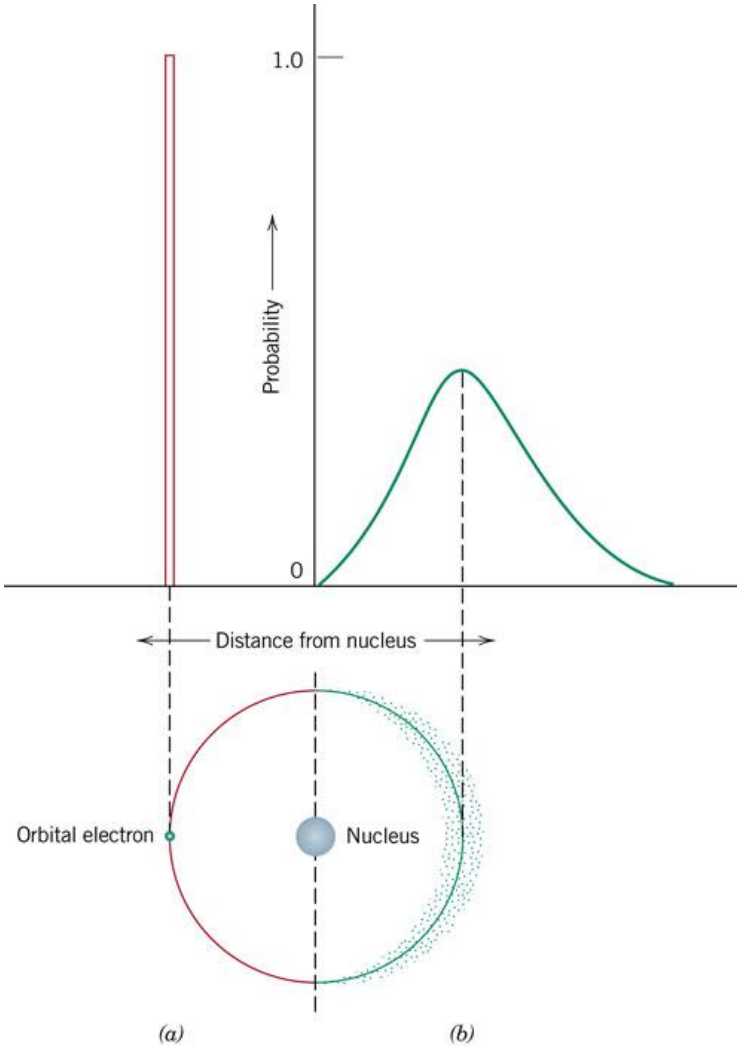
Reported atomic mass = abundance-weighted average of the isotopes, e.g. C: 0.989(12.000) + 0.011(13.003) ≈ 12.011 g/mol

Atomic Structure

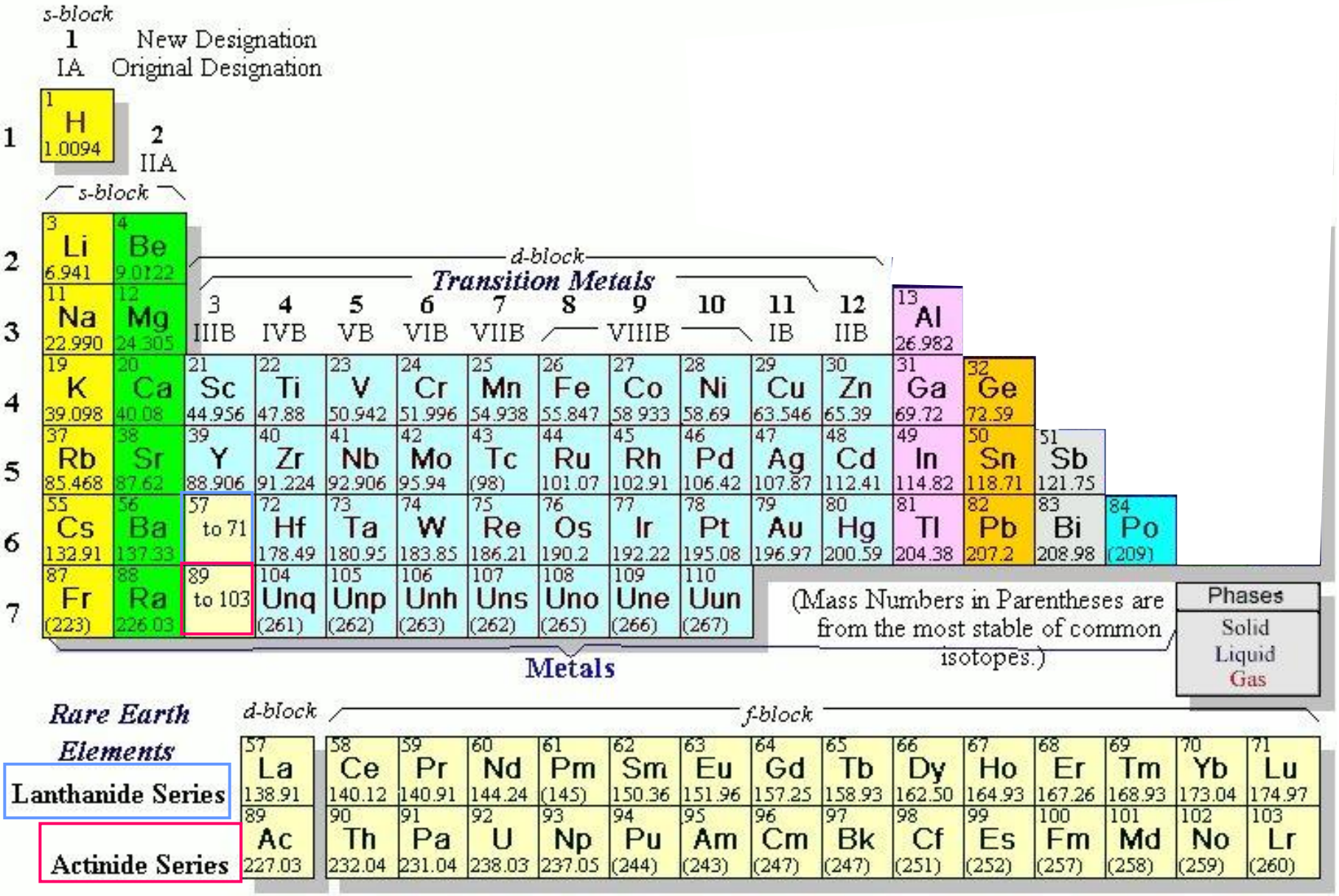
Bohr vs. Quantum-Mechanical Model: (1927)

- Wave-mechanical atomic model
  - Position of electron is imprecisely known; only a probability distribution.
  - Electron exhibits both particle and wave characteristics

Schroedinger, Heisenburg, Planck and others developed this model (wave mechanics), which allows a more precise description of the atom.



Increasing Electronegativity



The Periodic Table

Electronegativity - each kind of atom has a certain attraction for the electrons involved in a chemical bond. This "electron-attracting" power of each atom can be listed numerically on an electronegativity scale.

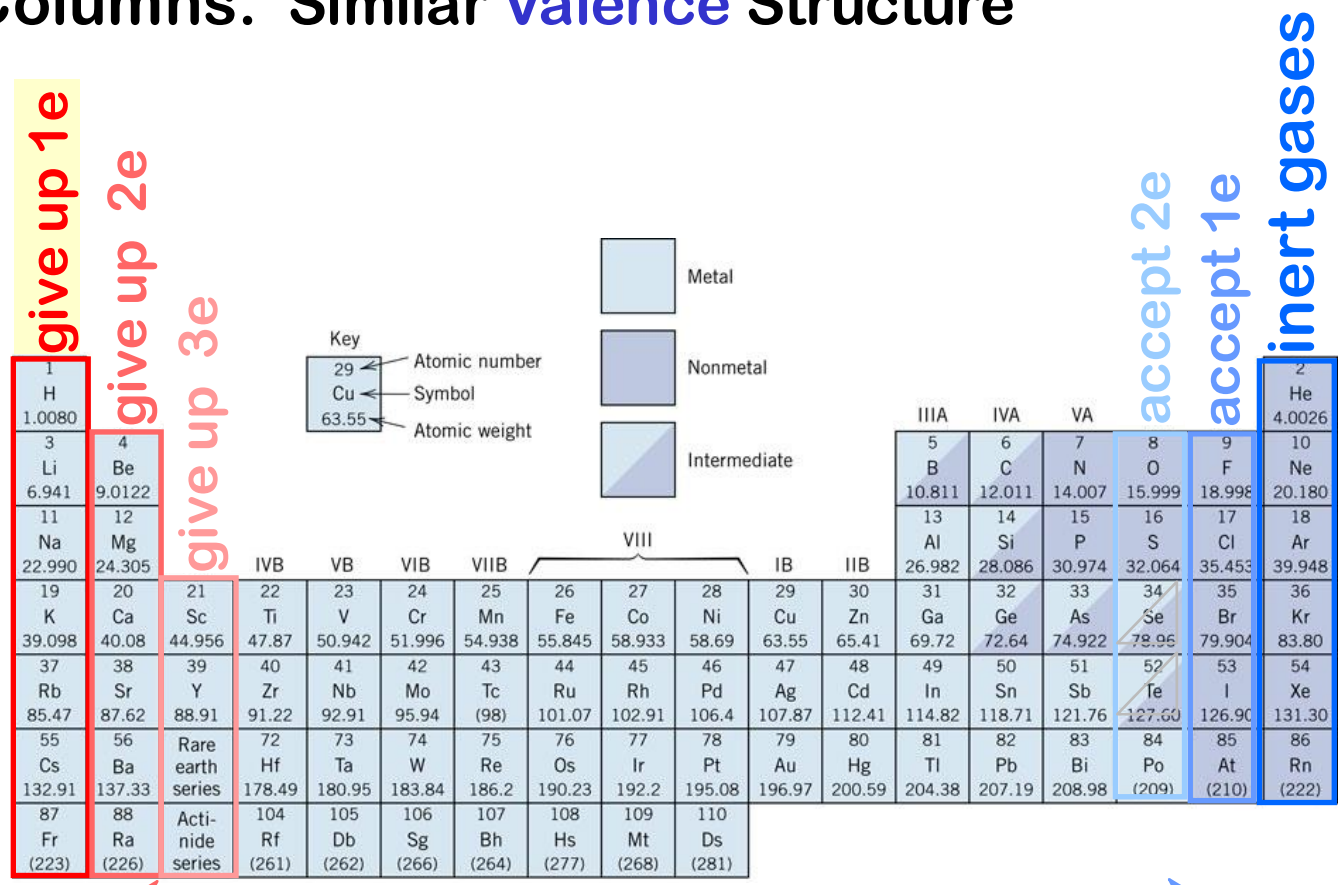
Electronegativity Values of Selected Elements

Metallic Elements			Nonmetallic Elements			
Li (1.0)	Be (1.5)		C (2.5)	N (3.0)	O (3.5)	F (4.0)
Na (1.0)	Mg (1.2)	Al (1.5)		P (2.1)	S (2.5)	Cl (3.0)
K (0.9)	Ca (1.0)	Sc (1.3)			Se (2.4)	Br (2.8)

Electronegativity was originally worked out by Linus Pauling in 1939 – see "The Nature of the Chemical Bond"

The Periodic Table

- **Columns: Similar Valence Structure**



Adapted from Fig. 2.6, *Callister 8e.*

Electropositive elements: Readily give up electrons to become + ions.

Electronegative elements: Readily acquire electrons to become - ions.

Electronegativity

- **Ranges from 0.7 to 4.0,**
- **Large values: tendency to acquire electrons.**

IA	IIA	IIIB	IVB	VB	VIB	VIIB	VIII			IB	IIB	IIIA	IVA	VA	VIA	VIIA	0
1 H 2.1																	2 He –
3 Li 1.0	4 Be 1.5											5 B 2.0	6 C 2.5	7 N 3.0	8 O 3.5	9 F 4.0	10 Ne –
11 Na 0.9	12 Mg 1.2											13 Al 1.5	14 Si 1.8	15 P 2.1	16 S 2.5	17 Cl 3.0	18 Ar –
19 K 0.8	20 Ca 1.0	21 Sc 1.3	22 Ti 1.5	23 V 1.6	24 Cr 1.6	25 Mn 1.5	26 Fe 1.8	27 Co 1.8	28 Ni 1.8	29 Cu 1.9	30 Zn 1.6	31 Ga 1.6	32 Ge 1.8	33 As 2.0	34 Se 2.4	35 Br 2.8	36 Kr –
37 Rb 0.8	38 Sr 1.0	39 Y 1.2	40 Zr 1.4	41 Nb 1.6	42 Mo 1.8	43 Tc 1.9	44 Ru 2.2	45 Rh 2.2	46 Pd 2.2	47 Ag 1.9	48 Cd 1.7	49 In 1.7	50 Sn 1.8	51 Sb 1.9	52 Te 2.1	53 I 2.5	54 Xe –
55 Cs 0.7	56 Ba 0.9	57–71 La–Lu 1.1–1.2	72 Hf 1.3	73 Ta 1.5	74 W 1.7	75 Re 1.9	76 Os 2.2	77 Ir 2.2	78 Pt 2.2	79 Au 2.4	80 Hg 1.9	81 Tl 1.8	82 Pb 1.8	83 Bi 1.9	84 Po 2.0	85 At 2.2	86 Rn –
87 Fr 0.7	88 Ra 0.9	89–102 Ac–No 1.1–1.7															

Smaller electronegativity ← → **Larger electronegativity**

Adapted from Fig. 2.7, *Callister 8e.* (Fig. 2.7 is adapted from Linus Pauling, *The Nature of the Chemical Bond*, 3rd edition, Copyright 1939 and 1940.

ELECTRONEGATIVITY

- High electronegativity → strong tendency to accept an electron (i.e., Group VIIA: F, Cl)

- Low electronegativity (called "electropositive") → strong tendency to give up an electron, i.e., Group IA: Li, Na, K)

The difference in electronegativity between two atoms determines the resulting *electron distribution* and the *type of bond*

Density, Atomic # & Wt, Mole & Avogadros

- Density
  - g/cm^3 (most solids range ~ 1 - 23 g/cm^3)
- Atomic number = number of protons (Z)
- Atomic weight (A)
  - g/mole

A ≅ number protons (Z) + neutrons (N)

≅ Z+N

- Mole = number of particles
  - N_A = …….. part/mole

Review Problems

- How many atoms in 6 grams of carbon?

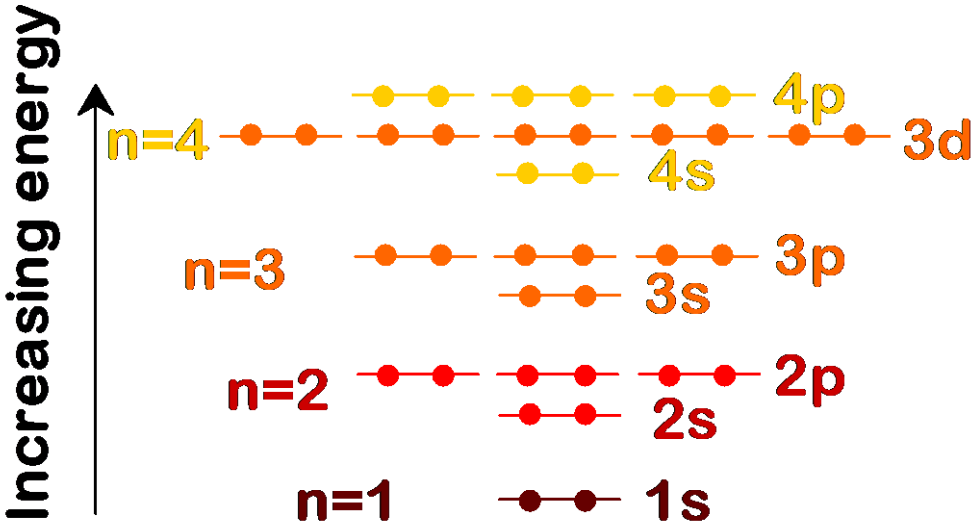
- Calculate the volume of 1 mole of Au.

Useful tip: many problems can be worked by suitable manipulation of density (g/cm^3), atomic mass (g/mole), and Avagadro's number (atoms/mole) (use dimensional analysis!)

Electron Energy States

Electrons...

- have discrete energy states
- tend to occupy lowest available energy state.

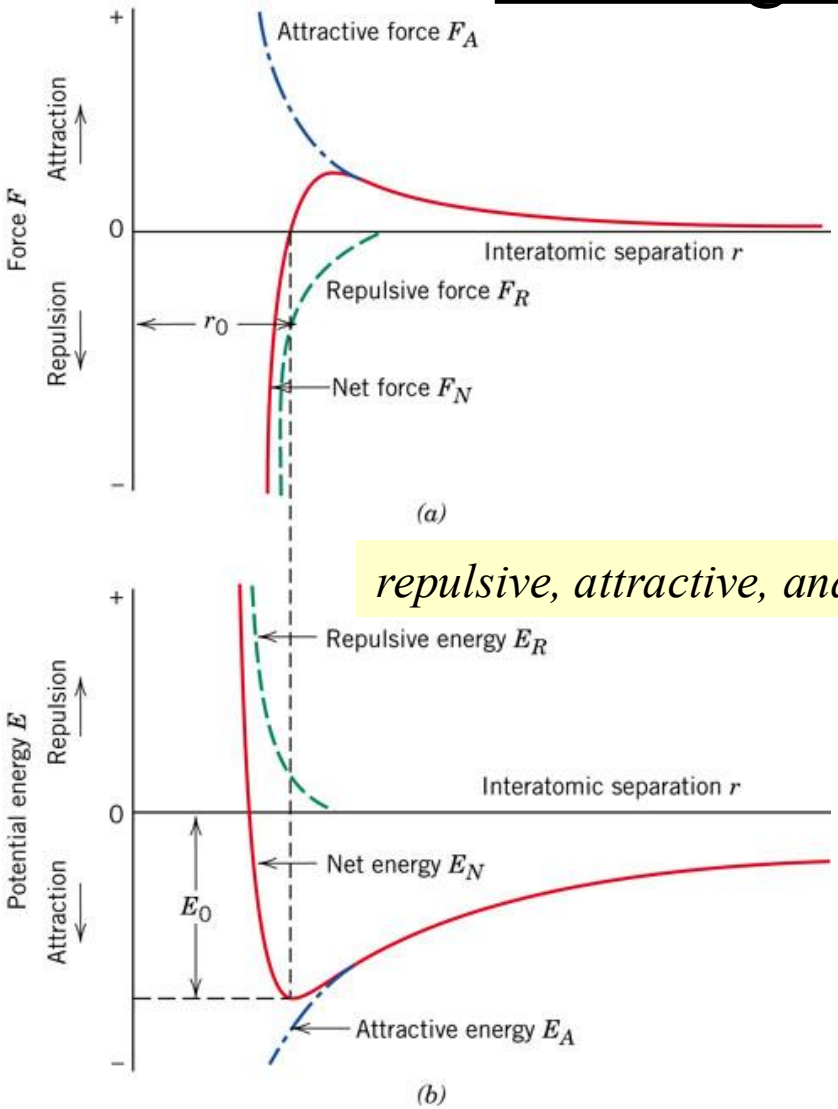


Stable electron configurations...

- have complete s and p subshells
- tend to be ……..

- Most elements: Electron configuration ……..

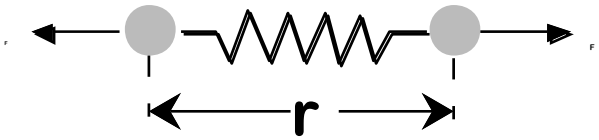
Bonding Forces and Energies



*repulsive, attractive, and net **forces***

*repulsive, attractive, and net **energies***

Bond length, r



Net force is given by the **sum** of an **attractive** force and a **repulsive** force

Potential is given by the integral of the net force curve with respect to distance:

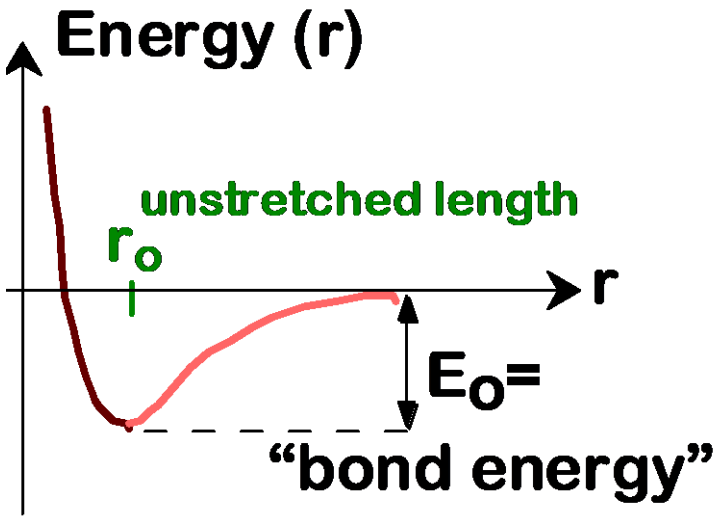
$$E = \int F \bullet dr$$

Note: *equilibrium separation occurs where the net force = 0*

Bonding Forces and Energies

Bonding energy: *Minimum of the potential vs. distance curve.*

- Indicates how much energy must be supplied to completely disassociate the two atoms
- Depth of the potential well indicates bonding strength
  - Deep well → ..... bonded
  - Shallow well → ..... bonded



Bond Energy

The higher the bond energy

- ……
- ……

- State as function of bonding energy
  - Solid (…….)
  - Liquid (……)
  - Gaseous (……..)

Next time:

Types of Atomic Bonds